A Literature Survey on Open-Source Turkish Text-to-Speech: Datasets, Models, and Emotion

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***Abstract*—Turkish is an agglutinative language spoken by over 83 million people [35], yet it remains underserved in text-to-speech (TTS) synthesis. This survey reviews 20 peer-reviewed papers, 10 community speech datasets, and 21 open-source Turkish TTS models. We evaluate each resource on three axes: dataset openness, model openness, and emotion support. Our findings show that no peer-reviewed work provides Turkish emotional speech data; the highest-quality system (MOS 4.39) is proprietary; 80% of HuggingFace datasets lack licenses; and community contributors have adopted modern architectures (F5-TTS, Orpheus) that have not appeared in academic Turkish TTS publications. We present these findings without prescriptive bias and let the evidence speak for itself.**

***Index Terms*—text-to-speech, Turkish, speech synthesis, low-resource, emotional speech, survey**

I. Introduction

Speech synthesis from text is used in screen readers, virtual assistants, customer service, and content creation. For high-resource languages like English and Mandarin, neural TTS systems now achieve near-human naturalness [22], [26], [27]. Languages with fewer digital resources have not kept pace.

Turkish has over 83 million native speakers and is an official language in Turkey and Cyprus [35]. Despite this, the open-source resources for Turkish TTS are sparse compared to languages of similar scale. In 2025, a Turkish robotic assistant still used eSpeak formant synthesis because no accessible neural TTS was available for offline use [13].

This survey catalogs Turkish TTS resources across three categories: (1) open speech datasets, evaluated for size, licensing, and documentation; (2) open TTS models, evaluated for architecture and community adoption; and (3) emotional speech resources. We do not advocate for any particular research direction. We present verified evidence and let readers assess whether the current landscape is sufficient.

A. Search Methodology

We searched Google Scholar, IEEE Xplore, Semantic Scholar, arXiv, DergiPark, and HuggingFace between February and April 2026 using both English and Turkish queries. From 22 initial papers and 25 HuggingFace repositories, we removed duplicates and unrelated entries, arriving at 20 unique papers, 10 datasets, and 21 Turkish-specific models. Each was individually verified.

II. Background

A. Speech Synthesis Methods

TTS has passed through four phases. Rule-based formant synthesis dominated the 1960s–1980s. Concatenative systems using unit selection followed in the 1990s–2000s [18]. Statistical parametric synthesis (SPSS) based on HMMs [33], [34] offered flexibility at the cost of naturalness. The neural era began with Tacotron [28] and WaveNet [29], followed by parallel and end-to-end architectures such as FastSpeech 2 [23], GlowTTS [30], VITS [22], and HiFi-GAN [24].

Since 2024, LLM-based TTS systems treat synthesis as token prediction. Orpheus [57] uses Llama 3B with a SNAC codec [59], Fish Speech [58] uses Qwen3 4B with a DAC codec, and VALL-E [31] introduced the paradigm. These systems support zero-shot cloning and, in some cases, emotion tags.

B. Turkish Linguistic Challenges

Turkish is agglutinative: suffixes attach sequentially to roots to encode grammar, tense, and possession [36], [37]. A single word like *görebilecekmiydik* (“were we going to be able to see?”) can replace an entire English clause. This produces a very large vocabulary and high out-of-vocabulary rates.

Although Turkish is often called phonetically transparent, Yilmaz [17] identified 154 exceptional syllables where grapheme-to-phoneme mapping breaks down. Heteronyms like *hala* (“aunt” vs. “still”), the silent ğ that lengthens vowels instead of producing a consonant, and circumflex vowels (â, î, û) all need contextual handling. Turkish abbreviations follow at least five reading conventions, from letter-by-letter (TBMM) to word-like (NATO) to foreign pronunciation (CD $\rightarrow$ *sidi*) [17].

Word stress is generally final-syllable, but exceptions are common for proper nouns and loanwords [39], [40]. Vowel harmony affects duration and formant patterns across syllables [38], [41].

C. Evaluation Metrics

The standard subjective metric is the Mean Opinion Score (MOS), rated on a 1–5 scale [42]. Objective alternatives include PESQ [43], STOI [44], MCD [46], and MOS-LQO via VisQOL [45]. Reference-free neural metrics such as MOSNet [47], DNSMOS [48], NISQA [49], and SSL-MOS [50] allow large-scale evaluation without human listeners.

III. Turkish TTS Systems

A. Concatenative and Rule-Based (2006–2012)

Sak et al. [18] built a corpus-based concatenative system at Boğaziçi University using Viterbi decoding and overlap-and-add waveform generation, reporting MOS 4.2. Tekindal and Arık [16] proposed Okuyucu TTS, storing only 386 syllable units and decomposing longer syllables at runtime. It achieved MOS 3.7 from 10 visually impaired evaluators. Its syllabification algorithm reached 100% accuracy on 10,000 Turkish words.

Yilmaz [17] built a text-processing front-end with 11 XML dictionaries for abbreviation expansion, liaison detection, stress, and heteronym handling. The identification of 154 exceptional syllables remains a useful reference for any Turkish G2P system. Later work by the same group added diphone synthesis with PSOLA [19] and rule-based prosody [20].

B. Neural Systems (2022–2023)

Ergun and Yildirim [5] explored cross-lingual transfer by remapping English phonemes to Turkish via Zemberek-NLP [60] and training FastSpeech 2 [23] on LJSpeech with Turkish labels. They reported no formal evaluation, noting the output sounded like "an English tourist trying to speak Turkish."

Cucen et al. [1] created a 12.6-hour balanced Turkish corpus and trained GlowTTS [30], achieving MOS-LQO 2.12. Oyucu [8] trained Tacotron 2 [25] + HiFi-GAN [24] on the same data, achieving MOS 4.49 from 40 evaluators and MOS-LQO 4.32 via VisQOL [45]. The dataset sits on Google Drive and the GitHub repo lacks HiFi-GAN code and pretrained weights.

Kocak and Büyükzincir [7] reported the highest Turkish TTS MOS at 4.39, using FastSpeech 2 + HiFi-GAN trained on 63 hours from a professional female voice actress. This Turkcell system is entirely proprietary.

C. Cross-Lingual Approaches

Yeshpanov et al. [6] built zero-shot TTS for 10 Turkic languages via IPA transliteration from Kazakh, using Tacotron 2 + Parallel WaveGAN [61] trained on 58 hours of Kazakh. Turkish received MOS 3.25 from 18 raters, with 91% comprehensibility and 61% intelligibility. The KazakhTTS2 dataset and models are openly available (CC-BY-4.0).

Karibayeva et al. [2] benchmarked MMS-TTS [56], TurkicTTS, ElevenLabs, and Coqui TTS for Turkish, finding all achieved MOSNet above 3.99. The 272,000-file corpus remains private.

D. Application Papers

Several papers use Turkish TTS without advancing it: a travel assistant [10] (MOS 2.98, nothing released), a robotic assistant using pyttsx3 [13] (no evaluation), a scene text reader [14] using Android TTS, and a banknote recognizer [15]. The reliance on eSpeak and platform TTS in 2025 highlights the accessibility gap.

IV. Open-Source Datasets

Ten unique Turkish speech datasets were found on HuggingFace. Table I summarizes them.

Only two declare a usable license: `afkfatih/turkish-tts-combined-raw` (declared CC-BY-SA-3.0, but effectively CC-BY-NC-SA-3.0 due to a mislabeled Khan Academy component) and `Anilosan15/Synthetic_Turkish_TTS_Data` (CC-BY-4.0, fully synthetic). The other eight have no license.

A large fraction of community data comes from YouTube extraction without permission. The falan42 series (13 repos, 18–25 hours) was extracted from monetized channels with up to 1.14 million subscribers. One dataset (`Appenlimited/700h-tr`) claims 700 hours but contains only 5.6 hours, a promotional sample from Appen's commercial product.

The largest aggregation (`afkfatih/turkish-tts-combined-ra` 81,513 samples, 120–140 hours) includes three MIT-licensed single-speaker corpora totaling roughly 60 hours, plus Khan Academy Turkish (75 hours, CC-BY-NC-SA-3.0) and Common Voice 17 Turkish (20 hours, CC0). It has no speaker IDs and no train/test splits.

For comparison, TatarTTS [3] provides 70 hours from two professional speakers under CC-BY-4.0, with open VITS models at MOS 4.54–4.65. Turkish, with 15 times more speakers than Tatar, lacks an equivalent.

V. Open-Source Models

Twenty-one Turkish-specific models were verified on HuggingFace. Table II lists them.

A. F5-TTS

Four Turkish F5-TTS [26] variants exist. Orkhon-TTS (Apache 2.0, alpha, 3.38 GB) supports voice cloning but has known issues with abbreviations. Two more by marduk-ra (CC-BY-NC-4.0) and Karayakar (MIT) were trained on Common Voice 17 Turkish. An ONNX conversion introduces a license conflict (CC-BY-4.0 from a CC-BY-NC-4.0 source).

B. Orpheus

The Orpheus architecture [57] uses Llama 3B [25] with SNAC [59] at 24 kHz. Turkish is not in the official release; all variants are community-built.

The Karayakar models (PT-2000 and PT-5000, MIT) were trained on 220+ hours and support emotion tags (`<laugh>`, `<sigh>`, `<gasp>`, `<cough>`, `<yawn>`, `<groan>`). These are the only Turkish TTS models with any emotion control. PT-5000 has 204 downloads/month. A GGUF version compresses the 13.2 GB model to 2.4 GB.

C. SpeechT5

Five Turkish SpeechT5 fine-tunes were found. The most downloaded (107/month) was an internship exercise that disclaims production use. One repo was completely empty. None report quality metrics.

D. Other Models

`facebook/mms-tts-tur` [56] is the most downloaded Turkish TTS overall (4,116/month), a VITS model with 36.3M parameters under CC-BY-NC-4.0. Kani-TTS (370M, LFM2 backbone) was trained on 61 hours of YouTube audio and restricts use to academic purposes. Karagoz-Hacivat-XTTS [62] is a culturally themed fine-tune for traditional shadow puppet theater. Only one Piper Turkish voice (dfki-medium, 63 MB ONNX) survives; two others were removed at contributor request.

E. Academic vs. Community Architectures

Academic papers use Tacotron 2, FastSpeech 2, and GlowTTS. Community contributors use F5-TTS, Orpheus, LLaSA, OuteTTS, and Dia. None of the 21 community models report MOS, WER, or any formal benchmark.

VI. Emotion in Turkish TTS

Emotional TTS is well-studied for English and Chinese, with datasets like ESD [51], EmoV-DB [52], and EMILIA [53]. Architectures such as Global Style Tokens [32] and emotion-conditioned models [54], [55] enable controllable synthesis.

For Turkish, the situation is different. Across all 20 papers in this survey, none address emotional speech. Every Turkish corpus in the literature was recorded with neutral delivery. Two papers [7], [10] mention emotional TTS as future work but do not implement it.

Among community models, the Karayakar Orpheus variants support non-verbal cue tags. This is not continuous emotion conditioning or categorical emotion selection; it is discrete insertion of laughter, sighs, and similar sounds. No community model supports arousal/valence or emotion categories (angry, sad, happy).

No Turkish emotional speech dataset exists in any form. This absence is categorical.

Multilingual models like XTTS-v2 [62] and Fish Speech [58] do produce Turkish speech with inherited expressiveness. For users who need "reasonably natural" Turkish output, these work today. The gap is in explicit, controllable emotion for applications like affective computing, audiobook production, or prosody research.

VII. Discussion

A. Reproducibility

No paper releases both an open dataset and a fully functional open model for Turkish TTS. Oyucu [8] comes closest, with a Google Drive dataset and GitHub code, but the HiFi-GAN vocoder is missing, no weights are confirmed, and the text normalization uses an English library with an empty abbreviation list.

B. Documentation

Eight of ten HuggingFace datasets have empty READMEs. Among models, 12 of 21 have at least minimal documentation, but only the Karayakar Orpheus models and the Karagoz-Hacivat model give full usage instructions.

C. Licensing

License metadata on HuggingFace cannot be trusted at face value. The largest dataset mislabels CC-BY-NC-SA-3.0 as CC-BY-SA-3.0. One model declares Apache 2.0 but restricts to academic use in its card. An ONNX conversion declares CC-BY-4.0 while deriving from CC-BY-NC-4.0.

D. Counterpoints

Turkish TTS is not entirely neglected. MMS-TTS-tur [56] gives instant lightweight synthesis. XTTS-v2 [62] enables voice cloning. The Orpheus community has 220+ hours of training data and emotion tags. The `afkfatih` aggregation provides 120–140 hours. The gaps are in formality: no benchmarks, no quality baselines, no clean licensing, and no emotional speech. For informal use, Turkish TTS works. For research and production, significant gaps remain.

VIII. Conclusion

This survey cataloged 20 papers, 10 datasets, and 21 models for Turkish TTS. Six findings stand out:

1) No Turkish emotional speech dataset exists. Two community models offer non-verbal cue tags.
2) The highest MOS (4.39) belongs to a closed Turkcell system.
3) Only 2 of 10 HuggingFace datasets declare a license; one is mislabeled.
4) 80% of datasets and 43% of models have empty documentation.
5) Community models use modern architectures; academic papers do not. Neither side provides formal benchmarks.
6) TatarTTS provides 70 hours with MOS 4.54–4.65 under CC-BY-4.0. Turkish, with 15x more speakers, has nothing comparable.

These findings point to opportunities in emotional speech data, reproducible pipelines, and systematic benchmarking. We leave the assessment of urgency to the reader.

Appendix: Summary Tables

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TABLE I: Open-Source Turkish Speech Datasets on HuggingFace

Dataset	Samples	Hours	kHz	License	Spk.	Split	Emo.	Notes
Appenlimited/700h-tr	2,000	5.6	16	Unknown	20	No	No	Mislabeled; commercial sample
Anilosan15/Turkish_TTS	30,606	33	48	None	1	No	No	YouTube scrape; copyright
mukahraman/orpheus-tr	1,000	–	24	None	–	No	No	Pre-tokenized; Orpheus only
falan42/Bentropi (11)	2,595	10	16	None	1	No	No	YouTube; copyright
falan42/Tunc_M	4,552	10	16	None	1	No	No	YouTube math lectures
falan42/Mert-H	837	3	16	None	1	No	No	YouTube; duplicated
afkfatih/combined-raw	81,513	130	24	CC-BY-SA-3.0*	100s	No	No	Largest; license error
afkfatih/snac-tokenized	81,513	130	24	CC-BY-NC-SA	100s	No	No	Orpheus only
Anilosan15/Synthetic	13,000	29	16	CC-BY-4.0	4	No	No	Synthetic; TTS unknown
yuserabv/turkish_tts	30,915	–	–	None	–	Yes	No	SpeechT5 features only

*Effective license is CC-BY-NC-SA-3.0 (Khan Academy component).

TABLE II: Verified Turkish-Specific TTS Models on HuggingFace

Model	Architecture	Size	License	DL/mo	Emo.	Notes
F5-TTS (Flow Matching + DiT)						
Orkhon-TTS	F5-TTS	3.4 GB	Apache 2.0	77	No	Alpha; voice cloning
marduk-ra/F5-Turkish	F5-TTS	4.1 GB	CC-BY-NC	–	No	3 checkpoints
Karayakar/F5-Turkish	F5-TTS	4.1 GB	MIT	–	No	Demo Space
Orpheus (Llama 3B + SNAC)						
Karayakar/Orpheus-PT-5000	Orpheus 3B	13.2 GB	MIT	204	Yes	Emotion tags; most popular
Karayakar/Orpheus-GGUF	Orpheus Q5	2.4 GB	MIT	69	Yes	CPU quantized
Cosmobillian/turkish_orpheus	Orpheus F16	6.6 GB	Apache 2.0	5	Yes	PT-5000 fine-tune
SpeechT5 (100–145M)						
Omarrran/speecht5_tts	SpeechT5	0.5 GB	MIT	107	No	Intern exercise
deryauysal/cv_tr	SpeechT5	0.6 GB	MIT	10	No	Common Voice
Other						
facebook/mms-tts-tur	VITS 36M	145 MB	CC-BY-NC	4,116	No	Most downloaded
Anilosan15/kani-tts-400m	LFM2 370M	740 MB	Apache 2.0	282	No	Academic only
SalihHub/karagoz-hacivat	XTTS-v2	3.8 GB	CC-BY-NC-SA	0	No	Cultural theme
Piper tr_TR-dfki	VITS ONNX	63 MB	MIT	–	No	Only Piper Turkish voice

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